

```

return 1;
}

int leap_year_calc(int year_lp)
{
const int ref_year=2012;
int lp_yr_flg=0;

if(year_lp<=2012)
{
if(((ref_year - year_lp)%4) == 0)
{
lp_yr_flg=1;
return lp_yr_flg;
}
}

else if(year_lp>2012)
{
if(((year_lp - ref_year)%4) == 0)
{
lp_yr_flg=1;
return lp_yr_flg;
}
}

return 0;
}

int no_of_days(int month, int year)
{
int day=0;
int leap_yr = 0;

leap_yr = leap_year_calc(year);

if((month == 1) || (month == 3) || (month == 5) || (month == 7) || (month == 8) || (month == 10) || (month == 12))
{
day=31;
return day;
}

else if((month == 4) || (month == 6) || (month == 9) || (month == 11))
{
day=30;
return day;
}

else if((month == 2) && (leap_yr == 1))
{
day=29;
return day;
}

else if((month == 2) && (leap_yr == 0))
{
day=28;
return day;
} return 0;
}

/*This function calculates the day when the date given by user is after the reference date*/
int aft_ref_date(int day_bef,int month_bef,int year_bef)
{
int day_dec,year_dec,month_dec=12,day=1;

for(year_dec=2012; year_dec<=year_bef; year_dec++)
{
for(month_dec=12; month_dec>=1; month_dec--)
{
for(day_dec=no_of_days(month_dec, year_dec); day_dec>=1; day_dec--)
{
if(year_dec==year_bef && month_dec==month_bef && day_dec==day_bef)
{
return day;
}

day--;

if(day==0)
{
day=7;
}
}
} return 0;
}

/*This Function calculates the day when the date given by user is after the reference date*/
int aft_ref_date(int day_aft,int month_aft,int year_aft)
{
int day_inc, year_inc, month_inc=1, day=1, month_day_lim;

for(year_inc=2012; year_inc<=year_aft; year_inc++)
{
for(month_inc=1; month_inc<=12; month_inc++)
{
month_day_lim = no_of_days(month_inc, year_inc);

for(day_inc=1; day_inc<=month_day_lim; day_inc++)
{
if(year_inc==year_aft && month_inc==month_aft && day_inc==day_aft)
{
return day;
}

day++;

if(day==8)
{
day=1;
}
}
} return 0;
}

void main()
{
int day, month, year, day_date, crct_date_flg=0;
const int ref_year = 2012;

printf("\nPlease enter the input as per the format\n\n");

printf("\nEnter the Date in DD format: ");
scanf("%d",&day);

printf("\nEnter the Month in MM format: ");
scanf("%d",&month);

printf("\nEnter the Year in YYYY format: ");
scanf("%d",&year);

crct_date_flg = correct_date_calc(day, month, year);

clrscr();
if(crct_date_flg == 0)
{
printf("Date Entered is an Invalid Date!");
getch();
}

clrscr();

if(crct_date_flg==1)
{
if(year <= ref_year)
{
day_date = bef_ref_date(day, month, year);
}

switch (day_date)
{
case 1:
printf("\nThe day was Monday");
break;
case 2:
printf("\nThe day was Tuesday");
break;
case 3:
printf("\nThe day was Wednesday");
break;
case 4:
printf("\nThe day was Thursday");
break;
case 5:
printf("\nThe day was Friday");
break;
case 6:
printf("\nThe day was Saturday");
break;
case 7:
printf("\nThe day was Sunday");
break;
}

else if (year > ref_year)
{
day_date = aft_ref_date(day, month, year);

switch (day_date)
{
case 1:
printf("\nThe day is Sunday");
break;
case 2:
printf("\nThe day is Monday");
break;
case 3:
printf("\nThe day is Tuesday");
break;
case 4:
printf("\nThe day is Wednesday");
break;
case 5:
printf("\nThe day is Thursday");
break;
case 6:
printf("\nThe day is Friday");
break;
case 7:
printf("\nThe day is Saturday");
break;
}

}

getch();
clrscr();
}
}

```